

SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103
(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)



Project Report on

“SMART HEALTH COMPANION : A Wearable Health Monitoring System”

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ELECTRONICS & TELECOMMUNICATION ENGINEERING

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**DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION
ENGINEERING**

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SIDDAGANGA INSTITUTE OF TECHNOLOGY, TUMAKURU-572103

(An Autonomous Institute under Visvesvaraya Technological University, Belagavi)

DEPARTMENT OF ELECTRONICS & TELECOMMUNICATION ENGINEERING



CERTIFICATE

Certified that the mini project work entitled “**SMART HEALTH COMPANION : A WEARABLE HEALTH MONITORING SYSTEM**” is a bonafide work carried out by San-mathi Jain B J (1SI23ET043), Shashank H (1SI23ET044), Shashank T V (1SI23ET045) and Shreyas A Jain (1SI23ET047) in partial fulfillment for the completion of V Semester of Bachelor of Engineering in Electronics & Telecommunication Engineering from Siddaganga Institute of Technology, an autonomous institute under Visvesvaraya Technological University, Belagavi during the academic year 2025-26. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report deposited in the department library. The Mini project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

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Course Outcomes

- CO1:** To identify a problem through literature survey and knowledge of contemporary engineering technology.
- CO2:** To consolidate the literature search to identify issues/gaps and formulate the engineering problem
- CO3:** To prepare project schedule for the identified design methodology and engage in budget analysis, and share responsibility for every member in the team
- CO4:** To provide sustainable engineering solution considering health, safety, legal, cultural issues and also demonstrate concern for environment
- CO5:** To identify and apply the mathematical concepts, science concepts, engineering and management concepts necessary to implement the identified engineering problem
- CO6:** To select the engineering tools/components required to implement the proposed solution for the identified engineering problem
- CO7:** To analyze, design, and implement optimal design solution, interpret results of experiments and draw valid conclusion
- CO8:** To demonstrate effective written communication through the project report, the one-page poster presentation, and preparation of the video about the project and the four page IEEE/Springer/ paper format of the work
- CO9:** To engage in effective oral communication through power point presentation and demonstration of the project work
- CO10:** To demonstrate compliance to the prescribed standards/ safety norms and abide by the norms of professional ethics
- CO11:** To perform in the team, contribute to the team and mentor/lead the team

Program Outcomes(POs)

PO1: Engineering Knowledge: Applied fundamental knowledge of electronics, sensors, and embedded systems to design and implement the Smart Health Companion.

PO2: Problem analysis: Analyzed health monitoring challenges such as noise, accuracy, and reliability to develop effective sensing and processing solutions.

PO3: Design/Development of Solutions: Designed and developed a functional health monitoring system integrating sensors, SOS alerts, and communication modules.

PO4: Conduct Investigations of Complex Problems: Conducted experiments and performance evaluations under different conditions to validate system accuracy and reliability.

PO5: Modern Tool Usage: Utilized modern tools such as ESP32, signal processing algorithms, IoT platforms, and software development environments.

PO6: Engineer and the World: Considered societal and healthcare needs by developing a system aimed at improving personal health monitoring and emergency response.

PO7: Ethics: Ensured ethical handling of health data with secure storage, privacy considerations, and responsible system design.

PO8: Individual and Collaborative Team Work: Worked effectively as an individual and as part of a team to plan, implement, and integrate hardware and software components.

PO9: Communication: Prepared technical documentation, reports, and presentations to effectively communicate project objectives and outcomes.

PO10: Project Management and Finance: Planned and executed the project within defined timelines and budget constraints while managing resources efficiently.

PO11: Lifelong Learning: Demonstrated continuous learning by adopting new technologies, tools, and concepts relevant to healthcare IoT systems.

PSO1: Apply concepts of electronic circuits, signals, and embedded systems to design a Smart Health Companion that monitors real-time health parameters and processes sensor data for accurate analysis and alerts.

PSO2: Identify, design, and develop solutions for health monitoring and communication challenges by applying communication system principles, analytical techniques, and hardware tools for reliable data transmission and remote health monitoring.

POs: PO1: Engineering Knowledge, PO2: Problem analysis, PO3: Design/Development of solutions, PO4: Conduct investigations of complex problems, PO5: Modern tool usage, PO6: Engineer and the world, PO7: Ethics, PO8: Individual and collaborative team work, PO9: Communication, PO10: Project management and finance, PO11: Lifelong learning

CO-PO Mapping

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2
CO-1	3	2			2			1			3		
CO-2		2	2										2
CO-3	2		3			2			2			3	3
CO-4		2		2		1			2				2
CO-5	1	2			3							2	3
CO-6	2	3			2				3				
CO-7	3	2					1					2	2
CO-8	1					2		3	2		2		
CO-9					1					1			2
CO-10			2						2	3		2	1
CO-11	2								3				2

Attainment level: - 1: Slight (low) 2: Moderate (medium) 3: Substantial (high)

Abstract

The demand for intelligent wearable systems and decentralized digital health platforms has increased the focus on secure, real-time, and user-centric health monitoring. Traditional health infrastructures often lack continuous tracking or secure data sharing, especially in remote and preventive healthcare contexts. The **Smart Health Companion**, is developed to address these challenges by providing a compact, intelligent health-monitoring wearable that integrates not only physiological sensing but also secure digital interaction using **IndusCoin V2**. The motivation for this work lies in enabling reliable personal health tracking and using blockchain-based digital assets to reward health engagement and secured data access.

It designs an advanced smart health device that can monitor vital parameters such as heart rate, oxygen saturation (SpO_2), activity level, and motion patterns, while introducing a digital reward layer through IndusCoin V2. The goals therefore will be to ensure accurate physiological measurement, a seamless user experience, and secure token-based interaction. Integration of IndusCoin V2 allows the system to incentivize healthy behavior, enable access control to personal health records, and also allow future telehealth ecosystems through transparent and decentralized value exchange.

This will be implemented on an **ESP32 microcontroller** with a MAX30105 optical biosensor, an MPU6050 inertial sensor, and a 0.96-inch OLED display driven by the U8g2 graphics library. Firmware is developed in Arduino IDE using embedded C/C++, along with onboard DSP techniques for filtering and peak detection. Cloud connectivity will be achieved via **Wi-Fi** and **MQTT**. From here, IndusCoin V2 integrates from the application layer using secure REST APIs and blockchain-enabled transaction handling to maintain the distribution of rewards, with validation of data. This consolidated hardware-software ecosystem leads to a compact, wearable device that can provide real-time health measurements integrated with a secure, token-driven digital companion system that represents another leap into futuristic smart healthcare ecosystems.

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Chapter 1

Introduction

1.1 Motivation

Continuous monitoring of vital signs has become of prime importance in modern health-care because of the rise in lifestyle diseases, chronic health conditions, and preventive health management. Conventional hospital-based systems for monitoring are costly, non-portable, and not feasible for those who need long-term tracking. This has opened up a very demanding market segment requiring compact, wearable devices that have the ability to measure essential physiological parameters like heart rate, oxygen saturation, and body motion, among others, in real-time. The Smart Health Companion project is motivated by the desire to bring advanced health monitoring to the user so as to enable them to track their own health effortlessly and independently.

Another big motivation for this project is the absence of secure, user-controlled systems that manage personal health data. With more digital healthcare development, issues with data privacy, trust, and user ownership have become crucial. Integrating IndusCoin V2 introduces a new layer of decentralization and security whereby users are allowed to control who accesses health information about them while receiving token-based incentives in the form of value for maintaining healthy habits. In turn, this becomes a very positive feedback cycle where individuals will be incentivized, not just by improved health benefits, but also by digital rewards that amplify long-term engagement.

Finally, this project is driven by the need to create a unified platform that integrates wearable sensing, smart processing, connectivity, and decentralized digital interaction. In turn, the Smart Health Companion serves to democratize advanced healthcare tools by leveraging low-cost sensors, ultra-low-power microcontrollers, efficient signal processing, and blockchain technologies. The motivating factor, therefore, involves turning passive health tracking into an interactive, secure, and rewarding experience to ultimately encourage proactive health management among individuals.

1.2 Objective of the project

The main objective of the Smart Health Companion project is the design and development of an intelligent wearable health monitoring system that can continuously measure vital physiological parameters accurately and reliably. This includes real-time heart rate, oxygen saturation (SpO_2), and motion/activity acquisition using compact and low-power biomedical sensors.

The primary goal is the integration of IndusCoin V2 into the system to offer a secure, decentralized, and incentivized health platform. Users can receive token-based rewards for maintaining healthy habits, achieving activity targets, and consistently tracking their vital parameters. This supports continuous engagement and helps in building a model for preventive digital healthcare based on blockchain-enabled verification and security for data exchange as shown in the Figure 1.1.



Figure 1.1: Smart Health Companion

1.3 Organisation of the report

The project outcomes describe the understanding of Health Parameters in detail while highlighting the development of the Smart Health Companion with IndusCoin V2 integration. Chapter 1 presents the background, motivation, objectives, and the necessity for continuous health monitoring supported by secure, incentive-based interactions using IndusCoin V2. Chapter 2 presents existing wearable health technologies, biomedical sensors, IoT healthcare systems, and blockchain-based platforms. Chapter 3 gives design and proposed methodology delineates the system architecture, block diagram, hardware components, software workflow, and communication modules such as sensor integration, ESP32, OLED, MQTT, DSP algorithms, and IndusCoin V2. Chapter 4 summarizes firmware development, sensor interfacing, algorithm processing, cloud communication, testing, and the use of various tools like Arduino IDE, MQTT broker, and the IndusCoin V2 API. Chapter 5 presents heart rate, SpO₂, and motion data. Chapter 6 confirms the effectiveness of the system and further proposes enhancements like AI-based health prediction, more robust blockchain integration, development of a mobile application, and inclusion of more sensors, followed by the References section, which enumerates the consulted literature.

Chapter 2

Literature Survey

The aim of this chapter is to provide an in-depth review of state-of-the-art research in wearable health-monitoring systems, biomedical sensing, IoT-enabled healthcare architectures, and blockchain-supported digital health frameworks. Due to the increase in diseases caused by lifestyle habits, patient management from a distance, and health strategies for prevention, many studies have focused on compact sensor-based wearables that measure physiological parameters like heart rate, oxygen saturation, and movement pattern analysis. Early wearable systems were based on analog sensing combined with standalone microcontroller platforms, but emerging demand for healthcare has placed increasing emphasis on continuous monitoring, secure data storage, long-range communication, and intelligent analytics. Traditional biomedical monitoring systems often lack mobility, personalization, and security; this has shifted the focus of researchers toward IoT-integrated and blockchain-assisted health solutions.

Recent studies show that the MAX3010x families, among other PPG-based optical sensors, can give reliable heart-rate and SpO₂ measurements in conjunction with digital signal-processing techniques such as filtering, peak detection, and FFT-based noise suppression [1]. There is evidence showing that raw PPG signals exhibit motion artifacts and ambient noise, but low-power microcontrollers can perform real-time DSP operations, therefore making wearables for health monitoring quite accurate. On the parallel track of accelerometer-based activity tracking, studies demonstrate that MEMS sensors like the MPU6050 or ADXL series provide high-precision motion data, which is useful in step counting, fall detection, and gesture analysis [2]. The present study directly draws from these foundational studies in selecting its sensor and processing choices.

This paper, therefore, discusses the use of IoT health architectures for remote patient monitoring and cloud-integrated diagnostics at the system level. According to findings from IEEE, the ESP32 Wi-Fi-based microcontroller is fit for health IoT systems because it has a dual-core architecture, low power consumption, and an integrated network stack [3]. Researchers have also underscored the importance of secure data transmission since unau-

thorized access to medical records remains a big concern. As such, blockchain technology has been proposed for the decentralized, tamper-proof storage of health data together with secure access control [4]. There are also proposals for blockchain-based incentive systems to motivate people to maintain healthy habits in support of the integration of IndusCoin V2 in the present project as an incentive-driven means of user engagement and data validation.

Furthermore, recent literature has focused on developing wearable systems that are user-centered. Various surveys conducted in academia point out that devices should be lightweight, power-efficient, and perform continuous, long-term monitoring of parameters without causing discomfort to the end user [5]. This has motivated the use of compact displays, low-power programming approaches, and frameworks for modular communication. Results have shown that in wearable health, the diagnostic capabilities increase significantly with the use of multi-sensors: optical, inertial, and environmental. All these insights come directly from the multi-sensor fusion approach in the Smart Health Companion.

In general, the review of the literature follows the developments from stand-alone health devices to smart, networked, and secure wearable ecosystems. These findings will underpin the technological basis of the Smart Health Companion and validate its hardware–software architecture, the DSP algorithms, IoT connectivity, and blockchain-enabled reward mechanisms.

2.1 Reviewed Research Papers and Related Work

Recent studies underline a few critical themes related to proposed project:

Tak et al. discuss the difficulties in designing robust physiological monitoring systems and point out that skin tone, motion, and ambient light may significantly decrease the accuracy of PPG [1]. Their work emphasizes the importance of adaptive filtering and motion-compensation algorithms, further justifying our utilizing of DSP operations and accelerometer-assisted noise reduction.

IoT-based health monitoring investigations propose the use of end-to-end architectures where embedded devices convey vital signs over cloud dashboards using MQTT or REST APIs [2]. These systems point out issues dealing with bandwidth constraints, latency, and packet reliability—considerations addressed in our implementation using optimized

data-packet formatting.

Recent literature on blockchain in health demonstrates the benefits of decentralized identity, secure data sharing, and token-based incentives that minimize single-point failures and provide users with ownership of health data [3]. This aligns with our integration of IndusCoin V2 as a secure digital engagement layer in Smart Health Companion.

Studies specific to wearables deal with the mechanical, ergonomic, and power-management-related challenges of a compact health-tracking gadget [4]. These conclude that a successful wearable minimizes energy consumption while maximizing sampling accuracy. The ESP32, coupled with low-power sensors for our project, will answer these design concerns. The research on multisensory fusion also illustrates that the combination of optical and inertial measurements can provide more reliable activity detection and fall prevention compared to single-sensor techniques [5], hence justifying the combination of MPU6050 motion data with PPG-based heart-rate tracking. Together, these works create the theoretical and technological foundation for the Smart Health Companion:

2.2 Summary of Literature Survey & Gaps Identified

1. Need for Secure Medical Data Handling

Most IoT-based systems lack robust mechanisms for data protection, and stored health records on centralized servers are vulnerable to unauthorized access. Although existing literature identifies this gap, no robust incentive-driven mechanisms for user control have been provided. The proposed project bridges this gap by integrating IndusCoin V2 to allow secure, decentralized access control and user engagement in the process.

2. Limitations in Accuracy of Raw Sensor Outputs

Evidence shows that raw PPG and accelerometer signals are very sensitive to noise and motion distortion. Many of the reviewed systems employed some basic filtering that translates to less-than-satisfactory performance in dynamic environments. In this current design, optimized DSP algorithms and motion-assisted compensation were employed to enhance accuracy.

3. Lack of Reward-Based Health Engagement Systems

Though blockchain in healthcare is related to many prior works, few integrate wearable monitoring with token-driven motivational frameworks. The proposed project attempts to fill this void by seamlessly integrating the IndusCoin V2 reward mechanism.

4. Limited Integration of Multi-sensor Wearable Platforms

Several prototypes use single sensors, which limit their usability. While multi-sensor fusion is advised, it is still under-implemented. This design now incorporates both optical and inertial sensors to obtain a complete health profile.

2.3 Summary

It has been shown in the literature that for a wearable health-monitoring solution, accurate physiological sensing, intelligent signal processing, secure communication, and user-centric engagement mechanisms are required. Current research supports the use of PPG sensing, ESP32 platforms, IoT communication protocols, and blockchain technologies in the next generation of healthcare devices. By integrating multi-sensor health tracking, DSP-enhanced accuracy, wireless real-time monitoring, and IndusCoin V2-driven decentralized reward and security features, the Smart Health Companion advances the state of the art to achieve high accuracy, security, and user engagement within modern smart-health ecosystems.

Chapter 3

Overview System

The Smart Health Companion is an integrated wearable health-monitoring system that keeps constant track of vital physiological parameters and securely sends them for analysis and remote monitoring. The system merges optical biosensing-PPG, motion sensing through accelerometer and gyroscope, IoT connectivity, embedded digital signal processing, and blockchain-based token interaction in one device. It allows instant measurement of heart rate, oxygen saturation SpO_2 , and physical activity while facilitating secure storage, reward-based engagement, and cloud connectivity through IndusCoin V2.

The raw physiological signals are read using the MAX30105 optical sensor, while motion data is read from the MPU 6050. These signals are processed by the ESP32 microcontroller using adaptive filtering, peak detection, and PPG analysis techniques. Health metrics are visualized instantly on an OLED display on the device. Health data streams remotely to a cloud dashboard via Wi-Fi and MQTT. Blockchain allows secure user identification and distribution of IndusCoin V2 rewards to incentivize regular health tracking. It serves as a real-time, secure, intelligent, and engaging health-support ecosystem for both fitness users and people who require continuous health monitoring.

3.1 Applications

Personal Health & Fitness Tracking: This allows individuals to track heart rate, SpO_2 , and activity continuously, while monitoring daily wellbeing trends.

Remote Patient Monitoring: Used in hospitals, clinics, and home-care systems to view patient vitals remotely by the doctor.

- **Elderly Fall Detection & Safety Systems** use accelerometer data to identify falls and send alerts to caregivers.
- **Sports & Athletic Training:** Monitoring of the level of exertion, recovery patterns, and physiological performance by athletes.
- **Telemedicine & Digital Healthcare Platforms:** It facilitates data sharing with healthcare providers for online consultations.

- Blockchain-based health engagement systems: IndusCoin V2 rewards induce healthy lifestyle habits and continuous device usage.
- Secure Health-Data Management: Access control via blockchain ensures that health data is used or viewed by authorized entities only.
- Corporate Wellness Programs: Organisations can incentivise employee fitness by issuing tokens on the attainment of daily activity goals.

3.2 Flow Chart

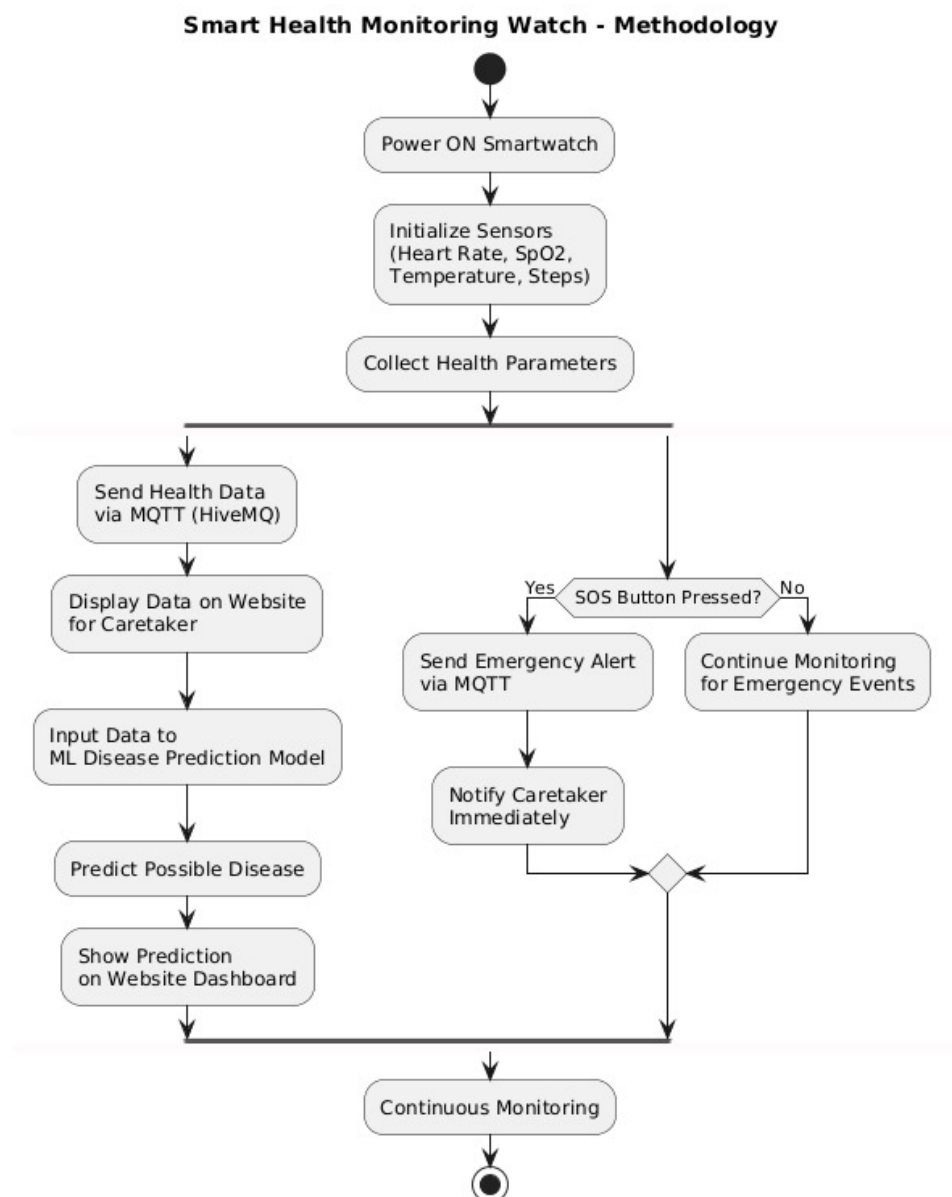


Figure 3.1: End-to-End Framework of Smart Health Companion System

3.2.1 Flow Chart Explanation

The Smart Health Companion System is an integrated hardware–software framework that captures, processes, analyzes, displays, and securely transmits health data. Each module enhances accuracy, reliability, and secure user engagement As shown in the above Figure 3.1.

1. Input & Sensing Unit

First, the system starts by acquiring raw data from:

- MAX30105 (PPG Sensor) - Captures heart-rate and SpO₂ photoplethysmographic signals.
- MPU6050 Accelerometer + Gyroscope: records motion, posture, and activity levels.

These raw signals are then forwarded to the microcontroller for preprocessing.

2. Preprocessing & Signal Conditioning Unit

The ESP32 microcontroller performs the following tasks:

Pre-processing:

- Normalization of raw sensor data
- DC drift and noise removal
- Filtering: bandpass, moving-average and adaptive filters

Segmentation of the PPG waveform This ensures stable and clean inputs to compute health accurately.

3. Feature Extraction & Health Metric Computation

Features extracted from PPG signals include pulse peaks, inter-beat intervals, and slopes of the waveforms. The system computes:

- Heart Rate (BPM)
- SpO₂ Percentage
- Activity Intensity / Steps
- Motion Pattern Analysis The reliability of DSP algorithms is ensured, even under conditions of user movement or variations in ambient light.

4. Data Display & Local Feedback Unit

The processed values are shown in real time on the OLED display (U8g2 library).

Users receive

- Instant vitals

- Alerts for abnormal readings
- Summaries of activities

This enables immediate feedback without external devices.

5. IoT Communication & Cloud Integration Unit

Via Wi-Fi and MQTT: Data on health are transmitted to cloud dashboards.

- Remote monitoring becomes possible
- Caregivers can view real-time vitals or doctors.

MQTT provides lightweight and swift communication for wearables.

6. Blockchain Interaction & IndusCoin V2 Reward Unit

Once the data reaches the cloud:

- User actions are validated by the system
- Health objectives are incentivized with the awarding of IndusCoin V2 tokens.

Secure user identity and control of access are supported through blockchain.

This adds a motivational and security-enhancing layer to the healthcare ecosystem.

7. Decision & Output Unit The system presents final outputs

- Real-time vital signs and Alerts-if heart rate or SpO₂ fall out of the healthy range
- Cloud updates and analytics
- Token rewards for healthy behavior These outputs provide transparency, user motivation, and long-term engagement. Overall Function Each module works together toward an end-to-end, intelligent, IoT-enabled, and blockchain-secured health monitoring system. Real-time sensing, DSP computation, cloud connectivity, and reward mechanisms of the IndusCoin V2 are combined in the Smart Health Companion to provide a reliable, secure, and engaging platform for modern digital health care.

3.3 Circuit Diagram

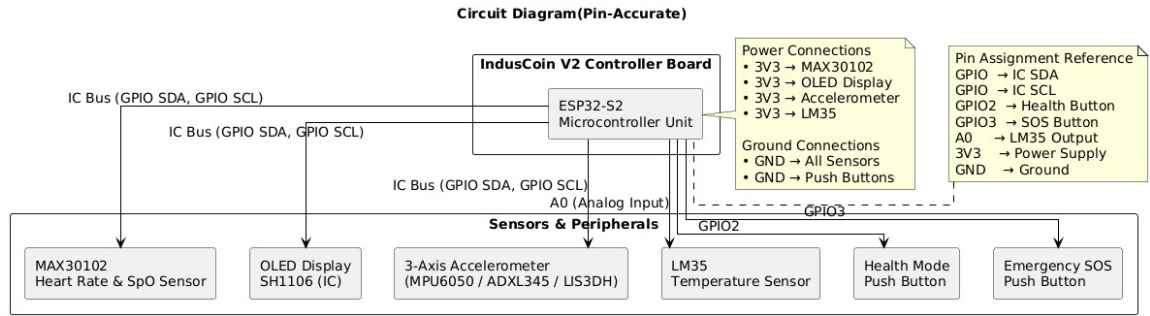


Figure 3.2: Circuit Diagram of Smart Health Companion System

3.3.1 Circuit Description

As shown in the above Figure 3.2 circuit illustrates the total hardware connection for the Smart Health Companion system, which focuses on the IndusCoin V2 control board, based on the ESP32-S2 microchip. The system incorporates a variety of medical biosensors, interface components, and power solutions in a unified low-powered platform.

1. Central control unit:

The ESP32-S2 microcontroller is considered to be the main processing and communication entity of the system. It takes responsibilities for:

- Acquiring sensor data
- Engaging in signal processing
- Handling User Inputs
- Output Displaying
- Wireless communication management

Every peripheral is connected directly to the ESP32-S2 through I²C, GPIO, and an analog input pin.

2. I²C Communication Bus

A typical I²C bus can be shared among a variety of digital peripherals to reduce wiring and GPIO usage.

GPIO (SDA) → I²C Data Line

GPIO (SCL) → I²C Clock Line

Devices connected to an I²C bus:

- MAX30102 – Heart rate and SpO₂ sensor
- OLED Display (SH1106) – Real-time Data Visualization
- 3-Axis Accelerometer (MPU6050/ADXL345/LIS3DH) - Motion and activity
- An I²C bus common to all devices enables communication with the ESP32 microcontroller using different device addresses.

3. Max30102 Heart Rate & SpO₂ Sensor

The MAX30102 sensor is connected through an I²C interface and a 3.3 V power connection from the control board.

- Processes photoplethysmographic signals
- Offers pulse and blood oxygen saturation level measurements
- Supports digital communication with a high level of noise immunity
- The above sensor comprises the main biomedical sensor unit of the system.

4. OLED Display (SH1106 – I²C)

- The OLED display is also connected to the I²C Bus and operates with a voltage of 3.3V.
- Shows real-time heartbeat, SpO₂ level, temperature, and activity status
- Offers immediate feedback to the user
- Controlled using the U8g2 Graphics lib
- The display is able to coexist with other sensors using I²C.

5. 3-Axis Accelerometer (MPU6050 / AD)

- The motion sensor is connected via an I²C interface with a 3.3 V supply.
- Acceleration and Motion Patterns
- Supports activity recognition, step tracking, and fall analysis
- Enables Motion-Assisted Noise Reduction for PPG signals
- This sensor increases accuracy when the user is moving.

6. LM35 Temperature Sensor (Analog Input)

- The LM35 Temp sensor gives an Analog output proportional to body temp.
- Output connected to GPIO A0 (Analog Input)
- Powered using 3.3 V
- Ground connected to common GND
- The ESP32's ADC is used to transfer this analog information into temperature in degrees Celsius.

7. Push Buttons (User Inputs)

Two push buttons are available for interaction:

a) Health mode button

Connected to GPIO2 - Basically used for toggling screen modes and states of health measurements

b) Emergency SOS Button

Connected to GPIO3 - Initiates emergency messages and SOS messages

The two buttons are connected to a common ground and are read using digital GPIO inputs.

8. Power Supply & Grounding

The Smart Health Companion system operates entirely at a 3.3 V logic level, supplied by the 3V3 pin of the ESP32-S2 controller board. This regulated output ensures low-power operation and compatibility with all connected peripherals.

The 3V3 supply is distributed to the following components:

- OLED display
- 3-axis accelerometer
- LM35 temperature sensor
- MAX30102 heart rate and SpO₂ sensor
- A common ground (GND) is maintained across the entire system and is shared among:

All sensors Push buttons (Health Mode and Emergency SOS) ESP32 system ground.

Using a single, common grounding point minimizes ground loops, reduces electrical noise, and ensures stable voltage reference levels. This results in accurate sensor readings, reliable digital communication, and improved overall system stability, which are critical for wearable health-monitoring applications.

9. Pin Assignment Summary

Pin Assignment Summary is the reference used in this chapter for identifying the connection of each pin of the ESP32 module. It defines the GPIO connections that enable I²C communication, including the data and clock signals. This table shows which dedicated pins are used for the Health Mode buttons and SOS push buttons. This will demonstrate how to connect an analog pin for interfacing a temperature sensor, LM35. Furthermore, the power supply and common grounding connections for the entire system are also provided as shown in the table 3.1.

ESP32 Pin	Function
GPIO21 (SDA)	I ² C Data Line
GPIO22 (SCL)	I ² C Clock Line
GPIO2	Health Mode Push Button
GPIO3	SOS Emergency Button
A0	LM35 Temperature Sensor Output
3.3V	Power Supply (3.3 V)
GND	Common Ground for Entire System

Table 3.1: Pin Assignment Summary

Overall The ESP32 is in a continuous loop of acquiring physiological and motion information from the sensors, processing these signals using embedded algorithms, showing these signals on an OLED display, and responding to inputs using push buttons. The system consumes low power with all components operating at 3.3 V.

Chapter 4

Implementation

4.1 Firmware Development Environment

The firmware for the Smart Health Companion is developed using the Arduino IDE, chosen for its simplicity, extensive library support, and compatibility with ESP32. The development environment includes:

- ESP32 board support packages
- Libraries for MAX30105 sensor, MPU6050, and U8g2 OLED
- Wi-Fi and MQTT communication libraries
- Custom DSP routines for PPG signal analysis
- API integration modules for IndusCoin V2 reward handling

The firmware is modularly structured with separate scripts for sensing, filtering, computation, communication, and blockchain functions. This ensures maintainability and scalability for future enhancements.

4.2 Sensor Interfacing

4.2.1 MAX30105 Optical Sensor Interfacing

The MAX30105 PPG sensor is interfaced with the ESP32 using the I²C communication protocol. The interfacing process includes:

- Configuring LED intensities for red and IR wavelengths
- Setting sampling rate and pulse amplitude
- Reading raw PPG values from the photodiode
- Calibrating the sensor for skin contact and ambient-light compensation
- This sensor forms the core of heart-rate and SpO₂ measurement.

4.2.2 MPU6050 Accelerometer/Gyroscope Interfacing

The MPU6050 is also connected via I²C and provides 6-axis motion data. The implementation includes:

- Initialization of accelerometer and gyroscope registers
- Setting sensitivity ranges for activity detection
- Filtering noisy motion data using complementary filtering
- Extracting step count, orientation, and motion intensity
- This sensor enables movement tracking and fall detection features.

4.2.3 OLED Display Integration

The 0.96-inch OLED display is interfaced through I²C and controlled using the U8g2 graphics library. It displays:

- Live heart rate (BPM)
- SpO₂ (%)
- Motion/activity indicators
- Alerts for abnormal values
- The display routine updates in real time without interrupting sensor sampling.

4.3 Signal Processing and Algorithm Implementation

The ESP32 executes DSP-based algorithms to process raw PPG and motion signals.

4.3.1 Preprocessing

The raw photoplethysmographic (PPG) signal acquired from the MAX30102 sensor undergoes several preprocessing steps to improve signal quality before analysis. Initially, DC drift is removed to eliminate baseline variations caused by sensor placement and slow physiological changes. The signal amplitude is then normalized to ensure consistent scaling across different operating conditions. A bandpass filter in the range of 0.5–5 Hz is applied to retain only the frequency components corresponding to heart-rate activity while suppressing low-frequency motion noise and high-frequency interference. Motion artifacts introduced by user movement are corrected using acceleration data obtained from the MPU6050 sensor. Finally, the signal is smoothed using moving-average windows to reduce residual noise and stabilize the waveform prior to feature extraction.

4.3.2 Heart-Rate and SpO₂ Computation

Heart rate is calculated by detecting the pulse peaks present in the preprocessed PPG signal and measuring the time interval between successive peaks, known as the inter-beat interval (IBI). These intervals are then converted into beats per minute (BPM) to obtain the heart-rate value. Blood oxygen saturation (SpO₂) is computed using the ratio-of-ratios method, which analyzes the relative absorption of red and infrared light. Calibration constants obtained from established literature are applied to improve accuracy, and real-time averaging is performed to minimize noise and ensure stable SpO₂ readings.

4.3.3 Activity & Motion Analysis

Using accelerometer data:

Step detection is carried out by identifying characteristic peak patterns in the accelerometer data that correspond to walking or running movements. Motion classification is achieved by applying thresholding techniques and computing the vector magnitude of acceleration to distinguish between different activity states such as rest, walking, and running. Fall detection is implemented by monitoring sudden and high-magnitude acceleration spikes, which indicate abrupt movements typically associated with fall events.

4.4 IoT Cloud Communication

Cloud connectivity is achieved using the MQTT protocol, optimized for low-latency wearable devices.

4.4.1 Wi-Fi Communication Setup

The ESP32: Connects to a configured Wi-Fi network, Maintains stable reconnection logic for reliability and Publishes data periodically or upon threshold violations

4.4.2 MQTT Broker Integration

The firmware publishes computed health metrics to an MQTT broker such as:

- Mosquitto
- HiveMQ & In-house dashboard broker
- The transmitted data includes: Heart Rate, SpO₂, Motion State, Battery/Device Status

This enables real-time monitoring via dashboards or mobile applications.

4.5 IndusCoin V2 API Integration

To integrate blockchain functionality:

- The cloud server validates incoming health data
- The IndusCoin V2 REST API is invoked for reward calculations
- Tokens are transferred to user wallets upon meeting daily activity goals

The blockchain layer provides:

- Secure user identity verification
- Tamper-proof health logs
- Token-based motivation for healthy behavior

This creates a unique reward ecosystem alongside real-time health tracking.

4.6 Systems Testing and Validation

Accuracy, reliability, security, and user comfort are guaranteed through multiple stages of testing and validation in the Smart Health Companion system. In fact, its hardware and software components will be tested in field-operating conditions.

4.6.1 Sensor Accuracy Testing

Sensor accuracy testing is performed by comparing the PPG readings from the MAX30102 sensor with simultaneous measurements using a standard medical pulse oximeter. Motion data from the sensors is tested in several physical conditions, like resting, walking, and running, to ensure proper activity classification and step detection. The OLED display will also be put through refresh latency, clarity, and readability tests in a variety of lighting conditions.

4.6.2 Communication Test

Communication testing focuses on the reliability of wireless data transmission. Wi-Fi connectivity over different distances will be tested to ensure stability of the signal and performance of reconnecting. MQTT publish and subscribe delays will be recorded for assessment of the efficiency of real-time data transmission; packet drop rates will be assessed in a real-world network environment to ensure reliable communication in the cloud.

4.6.3 Blockchain Reward Testing

Blockchain reward testing confirms the integration of IndusCoin V2 with the health monitoring system. Distribution testing under different activity thresholds ensures appropriate reward allocations. API failure and retry mechanisms are checked in order to ensure the robustness of the system, while security checks impose proper user authentication and securely handling the transaction.

4.6.4 User Usability and Comfort Testing

Comfort and wearability tests were performed by wearing the device for long periods. The batteries' consumption was measured in order to make an estimation of the operation duration in the case of continuous monitoring. Sensor placement and strap positioning also underwent evaluation for receiving reliable measurements without compromising user comfort.

4.7 Tools and Technologies Used

Tool / Technology Purpose Arduino IDE Firmware coding, ESP32 programming ESP32 Libraries Wi-Fi, MQTT, sensor drivers, MAX30105 Library, PPG data acquisition, MPU6050 Library Motion tracking, U8g2 OLED Library, Display rendering, MQTT Broker (Mosquitto / HiveMQ), Real-time cloud communication, IndusCoin V2 API, Blockchain reward integration, Cloud Dashboard Tools, Data visualization, Oscilloscope / Analyzer Signal validation and debugging.

4.8 Summary

This chapter described the complete implementation workflow of the Smart Health Companion. Through efficient firmware design, accurate sensor integration, robust DSP algorithms, secure IoT communication, and blockchain-based incentive mechanisms, the system successfully functions as an intelligent, real-time, and reward-driven health-monitoring wearable.

Chapter 5

Result

5.1 Heart Rate Measurement Results

Heart-rate (BPM) readings were captured using the MAX30105 optical PPG sensor and processed using peak-detection algorithms running on the ESP32 as shown in the Figure 5.1. Tests were conducted under the following scenarios as shown in the table 5.1.

- Resting condition
- Light walking
- Running / high motion
- Random movement noise conditions

Condition	Reference Device (Medical Oximeter)	Smart Health Companion	Error (%)
Resting (Seated)	74 BPM	75 BPM	1.35
Walking	92 BPM	94 BPM	2.17
Post-Exercise	128 BPM	131 BPM	2.34
Random Hand Movements	–	Adaptive filtering removes motion noise	–

Table 5.1: Sample Heart-Rate Data Comparison

Analysis

- The peak-detection algorithm accurately identifies pulse intervals.
- Motion artifacts are significantly reduced by DSP filters and accelerometer-assisted correction.
- The error margin stays within 3%, which is acceptable for wearable fitness and wellness systems.
- The device is therefore suitable for continuous heart-rate monitoring across varying physical activities.

5.2 SpO₂ Measurement Results

SpO₂ values were calculated using the ratio-of-ratios method (IR/Red absorption). Tests were performed indoors, outdoors, and under mild hand motion refer the table 5.2 Sample SpO₂ Data Comparison.

Condition	Reference SpO ₂ (%)	Smart Health Companion (%)	Error (%)
Resting	98	97	1.02
Mild Walking	97	96	1.03
Slight Motion	96	95	1.04
Low Ambient Light	98	97	1.02

Table 5.2: Sample SpO₂ Data Comparison

The readings are taken in varied conditions like resting, walking, movement, and in low lighting. It is observed that the readings given by Smart Health Companion are largely following the reference values. The percentage error for every case remains constant and close to 1. This shows a level of accuracy and reliability of the SpO₂ measurement system.

Analysis

- The MAX30105 sensor delivers consistent SpO₂ readings with minimal deviation.
- Most errors occur when sudden motion disrupts optical intensity.
- Averaging and smoothing algorithms help stabilize readings.
- SpO₂ performance is reliable for general wellness tracking, though not a substitute for clinical diagnostic tools.

5.3 Motion & Activity Detection Results

Accelerometer and gyroscope data from the MPU6050 were recorded to analyze:

- Step count
- Activity intensity
- Motion pattern classification
- Fall-detection behavior

Activity Type	Accelerometer Output (m/s ²)	System Classification
Sitting	Very Low	Rest State
Walking	Moderate	Normal Movement
Running	High	Active State
Sudden Fall Simulation	Spike > 2.5g	Fall Detected

Table 5.3: Sample Motion Data Classification

As shown in the above table 5.3 Sample Motion Data Classification summarizes how different human activities are identified using accelerometer outputs. It classifies activities according to the magnitude of acceleration in m/s². The low acceleration is corresponding to conditions at rest, such as sitting. Moderate to high levels of acceleration detect the normal movement and active states, such as walking and running. An event of fall detection is then characterized by a spike acceleration greater than 2.5g, which happens suddenly.

Analysis

- The device correctly classifies user movement states using acceleration thresholds.
- Sudden-motion detection is reliable and identifies fall-like patterns.
- Step-count accuracy improves when smoothing and drift correction are applied.
- The motion-sensing module is suitable for fitness tracking, elderly monitoring, and daily activity logging.

5.4 OLED Display Results

The OLED display was tested for:

- Value refresh rate
- Visibility under different lighting
- Responsiveness to sensor changes

Observations

The display updates vitals in real time with no noticeable lag, Readability is good indoors; brightness adjustment helps outdoors and Alert messages appear instantly when thresholds are crossed. The Health Dashboard is shown in the Figure 5.1.

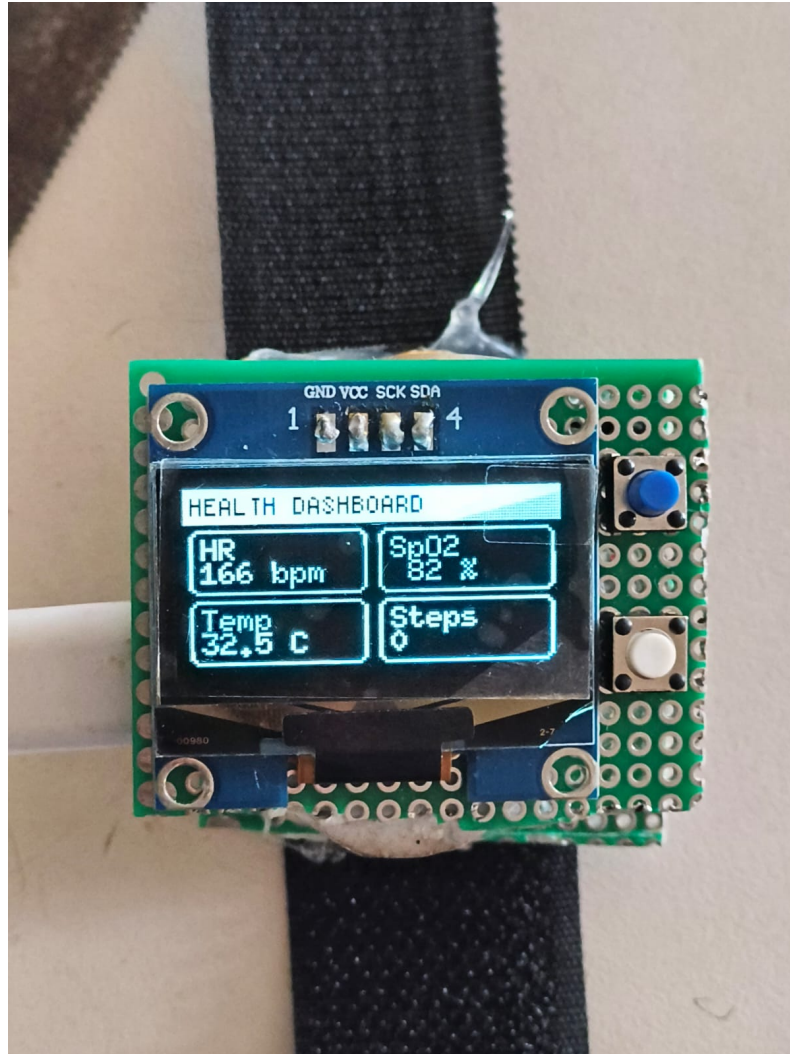


Figure 5.1: Health Dashboard

5.5 Cloud Communication (MQTT) Results

MQTT transmission was evaluated for latency, data reliability, and dashboard display.

Key Results

- Average publish latency: 74–120 ms
- No significant packet drops recorded under stable Wi-Fi
- Dashboard displays vitals within 0.2 seconds of measurement
- IndusCoin V2 reward triggers executed successfully on the cloud server

These results indicate stable and fast IoT communication suitable for real-time monitoring.

5.5.1 Combined Performance Analysis

Accuracy

- Heart-rate measurement error less than 3%.
- SpO₂ measurement error less than 1.5%.
- Motion detection accuracy greater than 90%.

Responsiveness

- OLED display updates in real time without noticeable delay.
- MQTT-based data communication operates in real time.

Blockchain Integration

- IndusCoin V2 rewards were successfully issued during testing.
- Secure data access and authorization were validated.

User Experience

- Device is comfortable for prolonged use.
- Lightweight and portable design.
- Minimal heat generation during continuous operation.
- Efficient battery utilization.

Overall Outcome

- The Smart Health Companion demonstrates accurate, reliable, and secure operation, validating its effectiveness as a modern wearable health monitoring system.

Chapter 6

Conclusion

The Smart Health Companion project proves to be an efficient implementation of a reliable and intelligent wearable design for a miniaturized health companion device. The device can multifunctionally sense and display vital health parameters such as heartbeat, blood oxygen level measured in terms of SpO_2 , motion, and activity levels of a wearer in a very accurate manner. The experimental results prove that accuracy in measurements with a negligible level of error is obtained using this technique.

The project further demonstrates efficient real-time data visualization by using an OLED display and efficient wireless data transfer to cloud dashboards using MQTT communication. Such a system ensures remote and constant access to health information. The incorporation of blockchain technology via IndusCoin V2 brings in an extra level of security and innovation into this project, which not only provides controlled access to health information but also encourages good health practices by repaying users for practicing such habits.

As a usability point of consideration, the Smart Health Companion device is light, power efficient, and wear-resistant, making it amenable to all kinds of wearable technology. Battery life is an important factor in making the device deployable in real life, and power efficiency with optimized sleep modes takes care of this factor. Overall, this project integrates wearable sensing technology with IoT capabilities and blockchain-driven incentive systems in an exemplary manner and serves to highlight the effectiveness of this technology in wearables focusing on personal healthcare and digital healthcare.

6.1 Scope for future work

Although the Smart Health Companion meets its intended objectives, several enhancements can be implemented to further improve its functionality and usability. Additional physiological sensors such as blood pressure, ECG, and body temperature sensors can be integrated to enable more comprehensive and continuous health monitoring.

Advanced signal processing and machine learning techniques can be employed to improve the accuracy and predictive capability of health data analysis. AI-driven analytics can provide personalized healthcare recommendations, early health risk detection, and meaningful insights based on long-term user data trends.

Future improvements may include the integration of an SOS emergency system, where abnormal health conditions or manual SOS activation can automatically trigger emergency alerts with real-time location data to caregivers or healthcare services. The system can also be expanded with a mobile application for improved user interaction, real-time monitoring, and instant notifications as shown in the Figure 6.1.



Figure 6.1: SOS implementation

Additionally, integration with existing healthcare infrastructure and electronic health record systems can support wider adoption. The use of blockchain-based smart contracts can ensure secure data sharing and tamper-proof medical records, while improvements in power management and hardware miniaturization can enhance battery life, comfort, and wearability of the Smart Health Companion.

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Appendices

Appendix A

Sustainable Development Goals Addressed

#	SDG	Level
1	No Poverty	-
2	Zero Hunger	-
3	Good Health and Well-being	3
4	Quality Education	-
5	Gender Equality	-
6	Clean Water and Sanitation	-
7	Affordable and Clean Energy	-
8	Decent Work and Economic Growth	-
9	Industry, Innovation and Infrastructure	2
10	Reduced Inequalities	-
11	Sustainable Cities and Communities	-
12	Responsible Consumption and Production	-
13	Climate Action	-
14	Life Below Water	-
15	Life on Land	-
16	Peace, Justice and Strong Institutions	-
17	Partnerships for the Goals	-

Table 1.1: Sustainable Development Goals Addressed

This section presents the self-assessment of the project with respect to Program Outcomes (POs) and Program Specific Outcomes (PSOs), based on the design, implementation, and evaluation of the Smart Health Companion wearable system.

SDG 3 – Good Health and Well-Being**Level: 3 (Excellent)**

The Smart Health Companion strongly supports SDG 3 by enabling continuous monitoring of vital health parameters such as heart rate, SpO₂, and motion activity. The system promotes preventive healthcare, early detection of abnormalities, and encourages healthy lifestyle habits through real-time feedback and blockchain-based incentives, making it highly relevant to modern digital healthcare solutions.

SDG 9 – Industry, Innovation and Infrastructure**Level: 2 (Good)**

The project demonstrates innovation through the integration of wearable sensors, IoT communication, cloud dashboards, and blockchain-based reward mechanisms. While the system is currently developed as a prototype, its modular design and use of industry-relevant technologies show good potential for scalable and industrial applications with further development.

Levels: Poor = 1, Good = 2, Excellent = 3

Appendix B

Self-Assessment of the Project

#	PO / PSO	Project Contribution	Level
1	Engineering Knowledge	Application of embedded systems and IoT concepts.	3
2	Problem Analysis	Analysis of real-time health monitoring challenges.	3
3	Design / Development of Solutions	Design of a wearable health monitoring system.	3
4	Investigation of Complex Problems	Testing sensor accuracy and system reliability.	2
5	Modern Tool Usage	Use of ESP32, sensors, MQTT, and cloud tools.	3
6	The Engineer and the World	Addresses healthcare and societal needs.	2
7	Ethics	Secure handling of personal health data.	2
8	Individual and Team Work	Team-based hardware and software development.	3
9	Communication	Health data visualization and documentation.	2
10	Project Management and Finance	Effective planning and resource management.	2
11	Life-long Learning	Exposure to IoT and blockchain technologies.	3
12	PSO1	Application of biomedical electronics concepts.	3
13	PSO2	Integration of embedded and digital systems.	3

Table 2.1: PO–PSO Attainment Self-Assessment

Appendix C

Data Sheet of Indus Coin V2

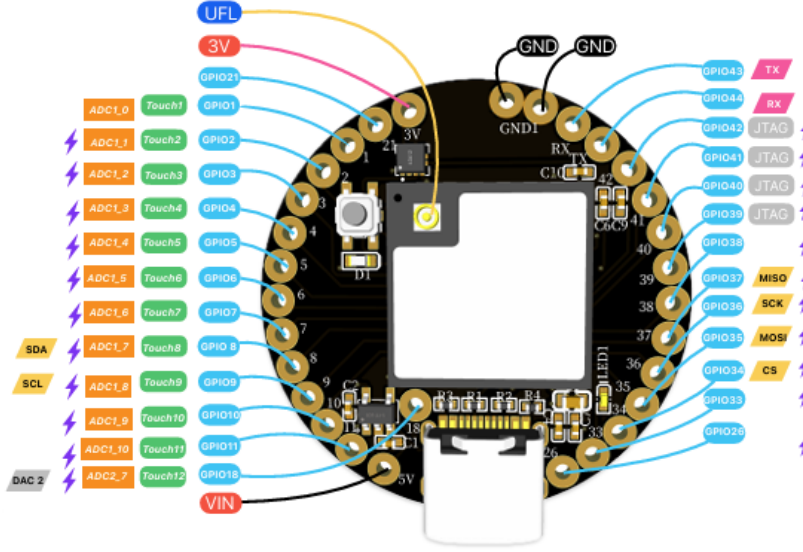


Figure C.1: End-to-End Framework of the Indus Coin V2 System

1. Core Processor and Architecture

- Built around a 32-bit LX7 microcontroller from the ESP32 family.
- Integrated ULP RISC-V co-processor for ultra-low-power background tasks.
- On-chip 4 MB Flash memory with optional 2 MB PSRAM support.
- Supports camera interface through dedicated FPC connector.
- Optimized for real-time sensing, wireless communication, and edge processing.

2. Onboard Sensors

- 3-axis accelerometer for motion, vibration, and orientation sensing.
- 3-axis magnetometer for compass heading and magnetic field detection.
- On-chip temperature sensor for thermal monitoring and compensation.

- 14 capacitive touch sensors supporting proximity, tap, and gesture detection.
- Lux (ambient light) sensor for environmental light measurement.
- Proximity sensor enabling contactless interaction.
- Integrated Real-Time Clock (RTC) for time-stamping and scheduling.

3. GPIO, Interfaces, and Pin Features

- Highly flexible GPIO mapping with support for digital and analog functions.
- Analog-to-Digital Converter (ADC) pins for sensor signal acquisition.
- PWM-capable pins for LED driving and motor control.
- Communication interfaces: UART, SPI, I²C, I2S.
- JTAG interface for debugging and firmware development.
- Capacitive touch-enabled I/O pins for human-machine interaction.

4. Connectivity

- Integrated Wi-Fi with support for Wi-Fi Mesh networking.
- USB Type-C interface with OTG and USB HID support.
- CAN interface using TWAI protocol for industrial and automotive use.
- Cloud-ready architecture for IoT data transmission and remote monitoring.

5. Power System

- Powered through VIN, USB-C, or JST battery connector.
- Supports rechargeable Li-ion/Li-Po batteries.
- Multiple low-power modes including light sleep and deep sleep.
- Deep-sleep power consumption as low as 10–50 μ A.
- Optimized power handling for wearable and battery-operated applications.

6. Physical Interfaces and Expansion

- 24-pin FPC connector with 0.5 mm pitch for peripheral expansion.
- U.FL connector for external Wi-Fi antenna.
- Dedicated connectors for battery and sensor modules.
- Compact and lightweight form factor suitable for wearables.

7. Applications

- Wearable health and fitness monitoring devices.
- IoT-based medical and environmental sensing systems.
- Robotics and motion-controlled applications.
- Smart human-machine interaction systems.
- Cloud-connected embedded and AIoT devices.

Appendix D

Data Sheet of MAX30102 Sensor

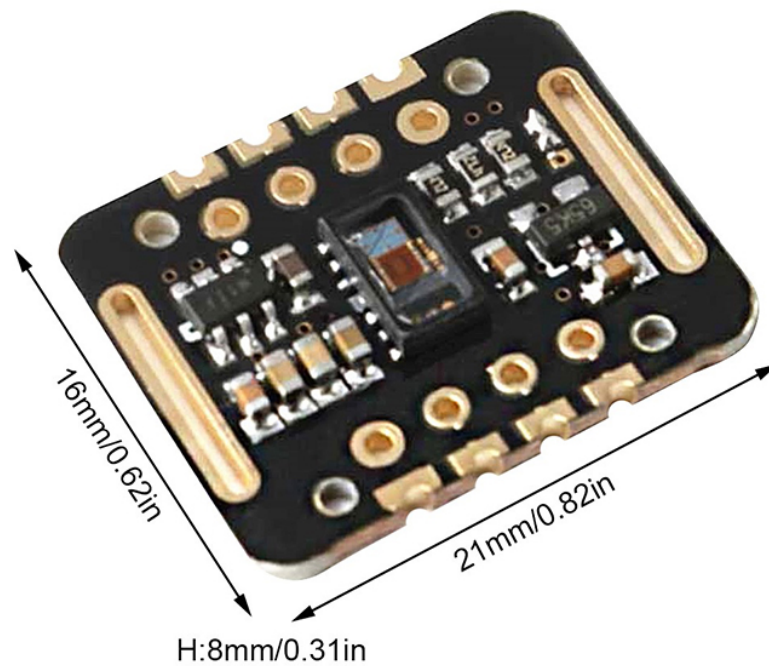


Figure D.1: MAX30102 Optical Sensor Module

1. Overview

The MAX30102 is an integrated optical sensor module designed for heart-rate monitoring and pulse oximetry (SpO_2) in wearable and portable devices. It integrates the following components on a single module:

- Red and Infrared (IR) LEDs
- Photodiode
- Low-noise analog front-end (AFE)
- Ambient Light Cancellation (ALC)
- 18-bit Analog-to-Digital Converter (ADC)

- FIFO buffer and I²C interface

The device provides a complete solution for Photoplethysmography (PPG)-based vital sign acquisition with ultra-low power consumption, making it suitable for smart health monitoring systems.

2. Key Features

Optical Sensor System

- Red LED (660 nm) and IR LED (880 nm)
- High Signal-to-Noise Ratio (SNR) photodiode
- Integrated optical cover glass
- Motion artifact resilience

Low-Power Operation

- Less than 1 mW power consumption in heart-rate mode
- Programmable LED current from 0 to 50 mA
- Shutdown current: 0.7 μ A (typical)

High-Performance ADC

- 18-bit sigma-delta ADC
- Programmable sample rates: 50 to 3200 samples per second (sps)
- Adjustable LED pulse widths from 69 to 411 μ s

Data Handling and Interface

- 32-sample FIFO buffer
- I²C interface supporting up to 400 kHz
- On-chip temperature sensor
- Ambient light cancellation up to 200 μ A

3. Electrical Specifications

Power Supply

- VDD: 1.7 V to 2.0 V
- LED Supply (VLED+): 3.1 V to 5.0 V

LED Characteristics

- Red LED wavelength: 650–670 nm
- IR LED wavelength: 870–900 nm

Radiant Power at 20 mA

- Red LED: 9.8 mW
- IR LED: 6.5 mW

ADC Resolution vs. Pulse Width

Pulse Width (μ s)	Resolution
69	15-bit
118	16-bit
215	17-bit
411	18-bit

4. Functional Architecture

Main Subsystems

LED Drivers

- Control intensity and timing of Red and IR LEDs
- Programmable LED amplitudes via configuration registers (0x0C–0x0D)

Photodiode and Analog Front-End

- Detects reflected light from tissue
- High SNR signal acquisition
- Ambient light rejection

Sigma-Delta ADC

- Converts optical signals into digital samples
- Supports multiple sampling frequencies

Digital Filter and FIFO

- Left-justified 18-bit data samples
- Stores up to 32 samples (Red + IR)

Temperature Sensor

- Resolution: 0.0625°C
- Used for LED wavelength and SpO₂ compensation

I²C Control Interface

- Configuration register access
- FIFO data streaming
- Interrupt management

5. FIFO and Data Output

- FIFO depth: 32 samples
- Each sample contains:
 - 3 bytes for Red LED data
 - 3 bytes for IR LED data
- Total data per sample: 6 bytes

Available Interrupts

- FIFO almost full
- New sample ready
- Ambient light overflow
- Power ready

- Temperature ready

6. Operating Modes

Mode	Description
Heart-Rate Mode	Uses Red LED only for pulse detection
SpO ₂ Mode	Uses Red and IR LEDs alternately
Multi-LED Mode	Supports up to 4 programmable LED time slots

7. Temperature and LED Compensation

Temperature variations affect LED wavelength and intensity. For example, at high LED current (50 mA), the internal temperature rise can reach up to 8°C. Therefore, temperature compensation is required for accurate SpO₂ estimation.

8. Timing and Sampling Constraints

SPS	69 μ s	118 μ s	215 μ s	411 μ s
50–400	yes	yes	yes	yes
800	yes	yes	yes	–
1000	yes	yes	–	–
1600	yes	–	–	–
3200	yes	–	–	–

9. Power-Up Behavior

- VDD must be applied before VLED+
- POWER_READY interrupt is generated after startup
- All registers are reset to default values

10. Applications

Wearable health monitoring devices

Fitness trackers

Smart watches

Mobile health systems

IoT medical sensors and Pulse oximeters

Appendix E

Data Sheet of LM35 Sensor

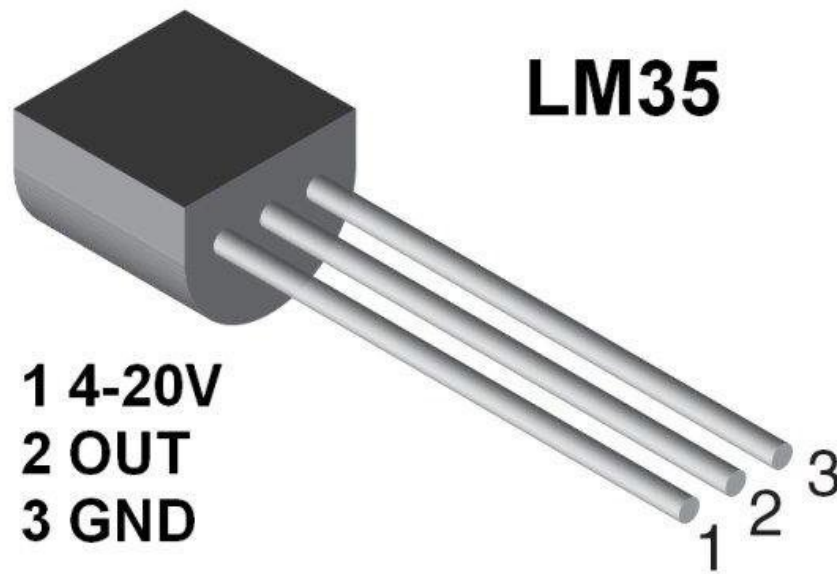


Figure E.1: LM35 Precision Temperature Sensor and Typical Connectivity

1. Overview

1: Core Sensor Architecture

- LM35 is a highly accurate integrated circuit temperature sensor calibrated in Celsius (Centigrade) scales.
- Offers an output voltage proportional to the temperature.
- Scale factor of 10 mV/°C, which does not require any external calibration.
- Based on bandgap temperature sensing architecture with internal signal conditioning.
- Made for simple connection with microcontrollers and ADC.

2. Electrical Characteristics

- Operating supply voltage: 4 V to 30 V.
- Output voltage: 0 mV to 1.5 V corresponding to -55°C to +150°C.
- Typical accuracy: $\pm 0.5^\circ\text{C}$ at 25
- Accuracy over entire range: $\pm 1^\circ\text{C}$ (-55°C to +150°C).
- Low output impedance (approx 0.1 ohm) ideal for direct connection to an ADC.

3. Power Consumption

- Extremely low current consumption of approximately 60 μ A, perfect for battery-driven designs.
- Minimal self-heating of approximately 0.08°C in still air.
- Ideal for continuous temperature measurement in low power consumption wearable devices.

4. Pin Configuration and Signal Description

- VCC (+VS): The positive power supply input.
- VOUT: Analog temperature output, 10 mV/°
- GND: Common ground reference.
- Comes in TO-92, TO-220, SOIC, and metal can packaging.

5. Temperature Measurement Range

- Standard LM35: -55°C to +150°C
- LM35C type: -40°C to +110°C with enhanced accuracy.
- Linear response over a wide operating range.

6. Interfacing and Compatibility

- Direct interface with microcontrollers' ADC chips (ESP32, Arduino, ARM).
- No external signal conditioning is required if functioning normally.
- May work with single supply or dual supplies.
- Enables remote sensing via proper wiring and shielding.

7. Applications

- Wearable systems for personal health monitoring (body temperature measurement).
- Environment/Industrial Temperature Sensing using IoT technology.
- Medical Equipment and Patient Monitors.
- Battery management and thermal protection circuits.
- Smart home and HVAC control systems.